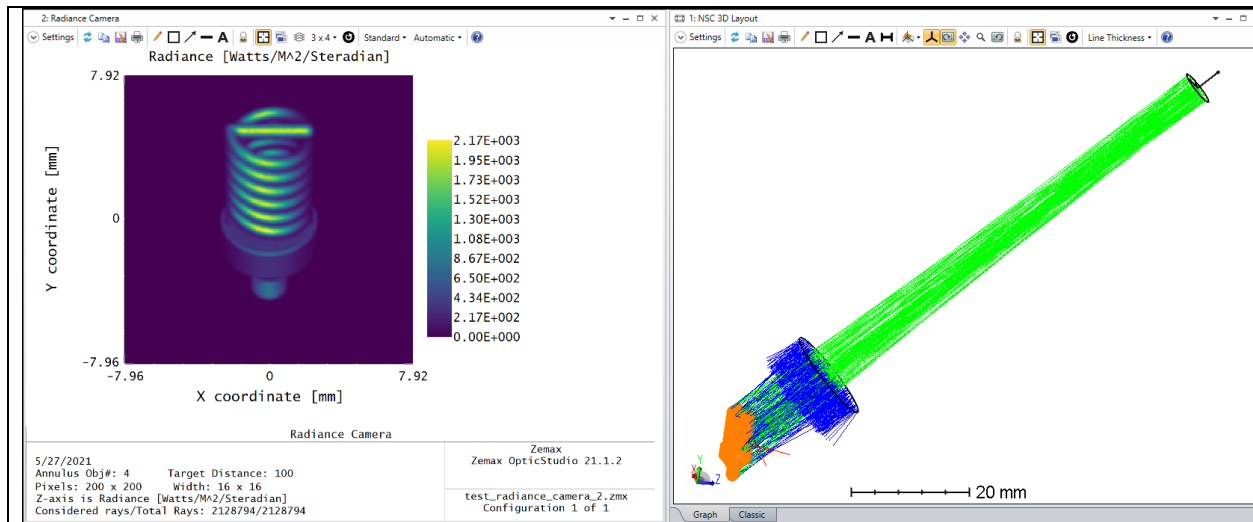


Title

API_Analysis_CS_Non-Sequential_Radiance_Camera

Picture



Overview

This tool calculate radiance/luminance on a distant plane seen by a human eye.

Code Exchange Type

API_Analysis

Language

CS

Author

Michael Cheng - Zemax

Updates

Date	Version	OpticStudio version	Comment
2021-05-27	1.0	21.2	Creation
2021-06-03	1.1	21.2	Add function to export text file
2021-06-11	1.2	21.2	Bug fix
2022-04-16	1.3	22.1	Bug fix when showing luminance data
2022-04-19	1.4	22.1	Bug fix. Now the extension file name is

Code Exchange

Zemax

			no restricted to lower case.
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Instructions

1. Installation

Open the ZAR example and the Radiance_Camera.exe will be automatically extracted to {Zemax}\ZOS-API\User Analysis.

2. How to use

0. This tool is not comprehensively tested via QA process and can be unstable. Contact Ansys Support if there is any questions.
1. 1. Setup an Annulus object in the system, which is your "eye pupi". Minimum X/Y Width should be always zero. Maximum X/Y Width should be same
2. Note this tool assumes rays are from -z and propagate in +z relative to the Annulus' local coordinate.
3. Now trace rays with saving rays hitting Annulus as source file. That means in Ray Trace Control dialog, you will set something like "##-abcd.sdf", where ## is Annulus object number, abcd is filename, .sdf is extension filename, which can be .dat instead.
4. Then go back to this tool's settings. X/Y Pixels/Widths are the size of the plane the eye is looking at. Target Z is the distance from the eye to this plane. Note this plane is always considered as at -z side of the Annulus object.
5. The tool detects setting in System Explorer > Unit > Source Unit and show luminance or radiance correspondingly.
6. Wave# is only used when needed. -1 means using primary wavelength. 0 means to use the wavelength data in the .dat file. Otherwise it's the number of system wavelengths.
7. If you check "Log", you can find a report in \Zemax\Objects\Sources\Source Files\radiance_camera.log. This is mainly for debug, but it's also useful if you want to check the source content.
8. The check box "Irradiance" is mainly for debug, but you could use it show irradiance or illuminance on the target plane.
9. Note you must retrace and save the source file whenever you changed the Annulus object's position or orientation. The tool detects if the rays included in the source file are on the Annulus and stop if they are not.